

Nilesh Amarathunge

• 408/9, Pinwalapitiya Watte, Bamunawala, Kurunegala.

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🌐 <https://portfolio.nilesh.uk/>

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Statement

I am a passionate and motivated individual with a strong interest in machine learning, robotics, and embedded systems. I am constantly seeking to expand my knowledge and skills through continuous learning, staying up to date with the latest advancements in technology to contribute to innovative and impactful solutions.



Education

University of Moratuwa

B.Sc. Eng. Hons. in Electronic and Telecommunication Engineering.

- SGPA semester 1 : **4.0(Dean's list)**
- SGPA semester 2 : **4.0(Dean's list)**
- SGPA semester 3 : **3.6913**
- CGPA : **3.8817**

Moratuwa, Sri Lanka

2024 - Present

National Institute of Business Management.

Diploma in Software Engineering.

- Distinction pass with GPA **3.911**

Kurunegala, Sri Lanka

2023 - 2024

Maliyadeva College.

GCE Advanced Level in physical science stream

- Combined Mathematics : **A**, Physics : **A**, Chemistry : **A**, General English : **A**
- Z-score : **+2.2132**

Kurunegala, Sri Lanka

2009 - 2023

Projects

Team N1K0 – Micromouse

2025 – Present

Micromouse Robotics Competition Project

Project link: portfolio.nilesh.uk

- Designed and manufactured a competitive micromouse for SLIIT Robofest 2025.
- Led full mechanical system design including chassis, drivetrain, and assembly.
- Designed a compact high-reduction (1:6) gear system for coreless motors.
- Performed full mechanical assembly and validation for smooth motor control.

Analog Noise Cancelling Headphones

2025 – 2026

Analog Electronics Product Design

Project link: portfolio.nilesh.uk

- Led a team to design and prototype an analog noise-cancelling headphone system.
- Designed complete analog circuitry including noise reduction and volume control.
- Created PCB layouts using Altium Designer and assembled the full prototype.
- Tested and validated circuit performance under real-world conditions.

Team Lynx – Battle Bot

2025 – Present

Combat Robotics Competition

Project link: portfolio.nilesh.uk

- Led a multidisciplinary team to design and manufacture a battle bot for RoboGames 2025.
- Designed a horizontal spinning blade weapon driven by a high-power DC motor.
- Completed full mechanical and structural design prior to manufacturing.
- Competed at RoboGames 2025 hosted by the University of Moratuwa.

Falcon E-racing

2025 – Present

Formula Student Competition

Project link: portfolio.nilesh.uk

- Falcon E-racing is the Formula Student group established in University of Moratuwa
- Volunteering at Falcon E-racing group as an low voltage engineer.
- Designed and manufactured the PCB's and wired the battery cart which handles charging the battery.

ThermoBand

2025 – 2026

Wearable Medical Device Product Design

Project link: portfolio.nilesh.uk

- Collaborated with electronics engineers to develop a consumer-ready wearable device.
- Designed enclosure optimized for comfort, durability, and manufacturability.
- Integrated sensors and electronics into final assembled product.
- Device enabled continuous body temperature monitoring and historical data tracking.

E-Commerce Web Application

2023 – 2024

Full-Stack Web Development Project

GitHub: [Frontend](#), [Backend](#)

- Led a team of three to design, implement, and deploy a full-stack e-commerce platform.
- Developed inventory management system for store employees.
- Implemented customer-facing features including browsing, ordering, and payments.

Skills

Courses

- **Supervised Machine Learning: Regression and Classification**
Completed through **Coursera**, with course content provided by **DeepLearning.AI** and **Stanford University**. This course covered the fundamentals of supervised learning, including regression, classification, model training, and evaluation techniques used in machine learning.
Date of Completion: 2026/03/16
Certificate: [View Certificate](#)

Languages

- Python (Proficient)
- C++ (Proficient)
- Java (Novice)

Other

- SolidWorks (Proficient)
- Altium Designer (Proficient)

Awards and Experience

- Robotic Minds Competition** 2015
1st place in National level Robotics Competition - World Robotic Olympiad(WRO).
- National Level Inventors Competition** 2014 - 2015
1st, 2nd and 3rd places in national and provincial level invention competitions.
• "Sahasak Nimavum", held by (Sri Lanka Inventors Commission) SLIC, and Ministry of Education, Sri Lanka.
- National Level Stamp Exhibition** 2017
1st place in the National Stamp Exhibition organized by the Philatelic Bureau of Sri Lanka.

Extra Curricular

- Student Representative** 2026-Present
A Student Representative of Department of Electronic and Telecommunications Engineering
- Member in Electronic club, University of Moratuwa.** 2025-Present
Web Developer under the Public Relations (PR) Branch of the E-Club, Department of Electronic and Telecommunication Engineering.
- Board member in Philatelic Society.** 2021
Vice Secretary of Philatelic Society, Maliyadeva College.
- Board member in Inventors Club.** 2021
Treasurer of Inventors Club, Maliyadeva College.
- Scout** 2012 - 2020
Active member of the Scout Troop at Maliyadeva College.

References

M. T. U. Sampath K. Perera

B.Sc. Eng. (Moratuwa, Sri Lanka), M.E.Sc. (Western, Canada), Ph.D. (RUB, Germany), MIEEE

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